

Reconciling Latent Variables and Networks

Exploring and extending the Psychometric-Toolbox

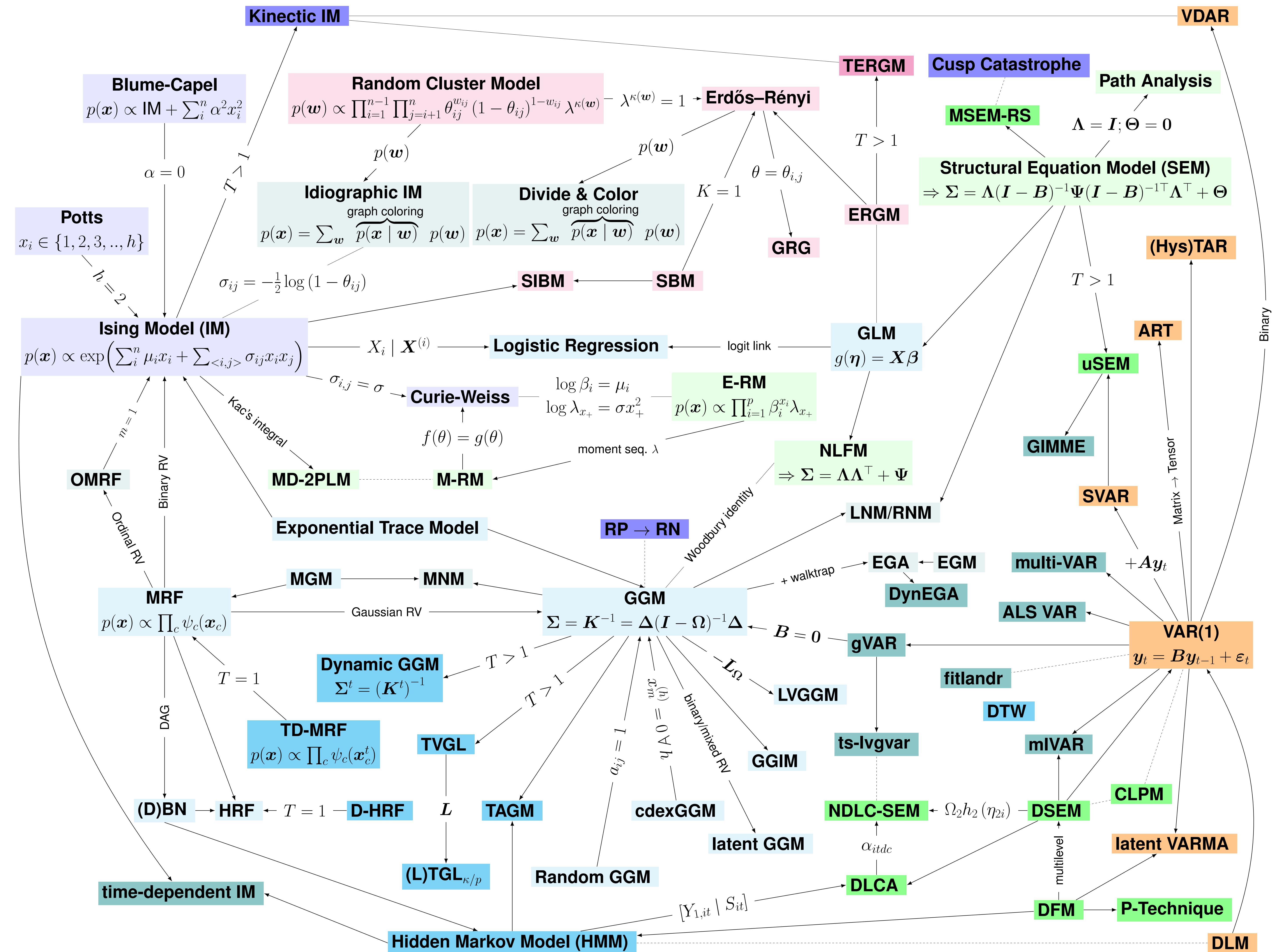


Figure 1: Areas of Research: Psychometrics/ (light blue), Network Psychometrics/ (light green), Econometrics, Statistics & Machine Learning/ (dark blue), Graph Theory & Network Science/ (pink), Physics/ (purple) (light colored for cross-sectional and darker colored for time-series data).

Models: Autoregressive Tensor Model (ART), Cross-Lagged Panel Model (CLPM), covariate-dependent GGM (cdexGGM), Dynamic Bayesian Network (DBN), Dynamic Factor Model (DFM), Dynamic Hybrid Random Field (D-HRF), Dynamic Linear Model (DLM), Dynamic Structural Equation Model (DSEM), Dynamic Time Warping (DTW), Erdős-Rényi (ER or Random Graph Model), Exploratory Graph Analysis (EGA), Exploratory Graph Model (EGM), Exponential Random Graph Model (ERGM), Extended Rasch Model (E-RM), Gaussian Graphical Interaction Model (GGIM), Gaussian Graphical Model (GGM), Generalized Linear Model (GLM), Generalized Random Graph (GRG), graphical VAR (gVAR), Group Iterative Multiple Model Estimation (GIMME), Hysteretic Threshold Autoregressive (HysTAR), Hybrid Random Field (HRF), Latent Network Modeling (LNM), Latent Variable Gaussian Graphical Models (LVGGM), Latent Variable Time-Varying Graphical Lasso (LTGL), Latent Variable Time-Varying Graphical Lasso with pattern inference (LTGL_p), Latent Variable Time-Varying Graphical Lasso with temporal kernel (LTGL_κ), Marginal Rasch Model (M-RM), Markov Random Field (MRF), Mixed Graphical Model (MGM), Mixture SEM with Regime-Switching (MSEM-RS), Moderated Network Model (MNM), Multidimensional 2-Parameter Logistic Model (MD-2PLM), Nonlinear Dynamic Latent Class - Structural Equation Modeling (NDLC-SEM), Ordinal Markov Random Field (OMRF), Residual Network Modeling (RNM), Structural Vector Autoregressive (SVAR), Stochastic Block Ising Model (SBIM), Stochastic Block Model (SBM), Temporal Exponential Random Graph Model (TERGM), Time Adaptive Gaussian Model (TAGM), Time-Dynamic Markov Random Field (TD-MRF), Time-Varying Graphical Lasso (TVGL), unified SEM (uSEM), Vector Autoregressive (VAR), Vector Discrete Autoregressive (VDAR)

